

Giovanni Madrigal, PhD

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EXPERIENCE

Postdoctoral Researcher

Sep. 2026 – Present

University of Kansas

Lawrence, KS

- Applying whole-genome approaches to characterize mito-nuclear interactions in Evolutionary Mutant Models as a framework for understanding human mitochondrial disease

Staff Scientist

May 2025 – May 2026

Frenzy Crop Systems Inc

Champaign, IL

- Directed genomic marker discovery efforts for agronomically important traits in major crop species, leveraging whole-genome sequencing data
- Served as sole cloud administrator for company AWS environment, establishing and maintaining EC2 compute instances, S3 storage for genomic data, and IAM user provisioning
- Developed custom bioinformatic tools and pipelines in Python, C++, and R to address analytical needs

Graduate Research Assistant

May 2021 – Aug. 2026

University of Illinois Urbana-Champaign

Champaign, IL

- Developed Klumpy, a Python-based tool for evaluating long-read genome assembly integrity, featuring novel algorithms for detecting sequence motifs with minimal dependencies
- Developed Synolog, a scalable C++ framework for synteny-based genome architecture characterization, incorporating custom visualization modules and algorithms designed for large-scale comparative genomics
- Identified a candidate gene underlying phenotypic variation through population-level genomic marker discovery
- Collaborated with 8 research groups across multiple departments and universities on projects spanning single-cell RNA-seq, eQTL mapping, genome assembly, and population genomics

Laboratory Technician

Jan. 2021 – May 2021

University of Miami Rosenstiel School of Marine and Atmospheric Science

Miami, FL

- Analyzed population structure and genetic diversity of an endangered species to support conservation management

TECHNICAL SKILLS

Languages: Python, C++, Bash, R, SQL, HTML/CSS

Developer Tools: Git, Conda, GDB

Cloud & HPC: Amazon Web Services (EC2, S3, IAM), Slurm

Computational Methods: Parallelization, data visualization, algorithm design

Genomics & Transcriptomics: Genome assembly, marker discovery, Single-cell and bulk RNA-Seq analysis

SELECTED PUBLICATIONS

Software

Madrigal, G., & Catchen, J. M. (2026). *Synolog: A scalable synteny-based framework for genome architecture characterization* [Under Review].

Madrigal, G., Minhas, B. F., & Catchen, J. (2025). Klumpy: A tool to evaluate the integrity of long-read genome assemblies and illusive sequence motifs. *Molecular ecology resources*, 25(1), e13982.

Applied Genomics

Benjamin, N. R., **Madrigal, G.**, Ishida, Y., Catchen, J., Allen, K. L., Pepin, B., & Roca, A. L. (2026). Genomic validation of perv-c-free pigs to support xenotransplantation. *Xenotransplantation*, 33(1), e70109.

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Ph.D. in Biology

May 2021 – Aug. 2026

Texas A&M University at Galveston

Galveston, TX

Master of Science in Marine Biology

Jan. 2018 – Dec. 2020

Texas A&M University at Galveston

Galveston, TX

Bachelor of Science in Marine Biology and Marine Fisheries

Aug. 2013 – Dec. 2017